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Paper 1: Immunostimulatory and Immunodynamic Modeling Analysis to Determine a Plausible S

Bibliographic Information

- **Title**: Immunostimulatory and Immunodynamic Modeling Analysis to Determine a Plausible Starting Dose of mRNA-4359 for Use in First-In-Human Trials
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Quick Take

Pharmacometricians working on early-phase mRNA immunotherapy development will find this paper essential, as it demonstrates how a semi-mechanistic IS/ID model can bridge between a peptide vaccine and an mRNA platform to support first-in-human (FIH) dose selection without preclinical animal data. The analysis shows that a 100 µg starting dose of mRNA-4359 is predicted to lie near the inflection point of the dose–response curve, balancing therapeutic potential with safety. This work is a pragmatic example of model-informed drug development (MIDD) applied to a novel immunotherapy modality.

Executive Summary

This paper applies a modified immunostimulatory/immunodynamic (IS/ID) model to determine a first-in-human (FIH) starting dose for mRNA-4359, a lipid nanoparticle–encapsulated mRNA immunotherapy encoding IDO and PD-L1 antigens. In the absence of preclinical animal data, the model was calibrated to digitized clinical data from a phase 1/2 study of an IDO/PD-L1 peptide vaccine in patients with metastatic melanoma (NCT03047928). Dose equivalence calculations and sigmoidal dose–response assumptions predicted that 180 µg of mRNA-4359 would elicit a T-cell response similar to 200 µg of the peptide vaccine, with a plausible range of 45–360 µg based on potential efficiency differences. Model simulations further predicted that a 100 µg dose would be appropriate as the FIH starting dose, ensuring patients receive potentially therapeutic doses while maintaining safety. This work represents a novel application of semi-mechanistic immune modeling to accelerate mRNA immunotherapy development.

Scientific Context & Motivation

mRNA-4359 is a novel LNP-encapsulated mRNA immunotherapy encoding two tumor-associated antigens (IDO and PD-L1). Because of the absence of an appropriate animal model that recapitulates human T-cell responses, conventional allometric scaling or toxicology-based approaches for FIH dose selection were not feasible. The peptide vaccine MM1636 trial provided the only clinical data with a similar mechanism of action (activating IDO- and PD-L1-specific T cells). Thus, the key challenge was to translate dose and immune response from a peptide-based platform to an mRNA platform using mechanistic modeling. This study fills a gap by providing a framework for dose selection in the absence of preclinical in vivo efficacy data, a common scenario for immuno-oncology agents targeting human-specific MHC-peptide complexes.

⚡ Methodological Snapshot

The authors adapted a semi-mechanistic IS/ID model (originally developed for a tuberculosis vaccine) to describe the kinetics of IFN-γ-secreting T cells following mRNA-4359 administration. The model includes two T-cell subtypes (TEM and CM) with dose-dependent activation and conversion functions. It was calibrated to digitized mean data from a peptide vaccine trial (NCT03047928) stratified by antigen and clinical response. Dose equivalence between the peptide vaccine and mRNA-4359 was calculated based on molecular weights and protein expression yields. Sigmoidal dose–response relationships were assumed based on prior mRNA platform data. The model was then used to simulate T-cell responses across a range of doses (1–1000 μg) and three dosing regimens (Q2W, Q3W, hybrid), leading to the recommendation of a 100 μg starting dose and a 15-cycle Q3W regimen.

Structural Model Breakdown

The IS/ID model consists of two ordinary differential equations (ODEs) describing the dynamics of transitional effector memory (TEM) and central memory (CM) T cells in peripheral blood. TEM cells are generated by an activation function $\delta(t)$ that is dose-dependent (sigmoidal E_{max} model). They are converted to CM cells at rate β_{TEM} (also dose-dependent) and die at rate μ_{TEM}. CM cells, upon re-treatment, can replicate at rate R_{CM} for a period τ and convert back to TEM cells at rate β_{CM}. CM cell death is governed by μ_{CM}. The model includes a cap of 1000 SFU/million PBMCs for total T cells. Separate parameter sets are used for IDO vs. PD-L1 antigen-specific responses and for CR vs. PR patient subgroups. The initial condition assumes zero TEM cells (no prior vaccination) and a small baseline of CM cells (not specified). The key time-varying input is the re-treatment schedule (doses), and the activation function δ and conversion rate β_{TEM} are modulated by dose magnitude via sigmoidal functions.

Detailed Methodological Analysis

Modeling Approach

Semi-mechanistic IS/ID model with two compartments (TEM and CM T cells), adapted from a previously published tuberculosis vaccine model. Key modifications include: zero baseline TEM (unvaccinated subjects), dose-dependent activation function (δ) and TEM-to-CM conversion rate (β_{TEM}) via sigmoidal E_{max} models, and simultaneous CM replication and conversion upon re-treatment. Total T cells capped at 1000 SFU/million PBMCs. Separate parameterizations for IDO vs. PD-L1 and CR vs. PR. Implemented in MATLAB.

Data Sources

Digitized mean T-cell response (IFN-γ ELISpot, SFU/million PBMCs) from the phase 1/2 MM1636 trial (NCT03047928) evaluating an IDO/PD-L1 peptide vaccine combined with nivolumab in 30 patients with metastatic melanoma. Data were stratified by antigen (IDO vs. PD-L1) and by RECIST 1.1 response (CR vs. PR). PD patients were excluded due to lack of sustained T-cell responses and potential confounding immune escape mechanisms.

Estimation Methods

Parameter estimation combined heuristic fixing of some parameters (based on literature or prior knowledge) with optimization algorithms for the remaining parameters. No automated estimation of all parameters was possible due to limited data points. Parameters defining the sigmoidal dose–response relationships (a, β_{TEM}) were guided by prior mRNA platform knowledge and adjusted for qualitative consistency.

Model Evaluation

Model fit was evaluated visually by overlay of predicted mean trajectories on observed digitized data (Figure 2). A global sensitivity analysis using the Sobol method was performed, drawing 1,000 parameter values uniformly from 0.5× to 2× baseline values, indicating that all estimated parameters influenced the output and that time of re-treatment (TR) was particularly sensitive. No VPC or bootstrap was conducted.

Covariate Analysis

No formal covariate analysis was performed; instead, data were stratified by RECIST 1.1 response (CR vs. PR) and antigen type (IDO vs. PD-L1) based on observed differences in T-cell response magnitude. This stratification was justified by the small number of patients and the availability of subgroup-level mean trajectories rather than individual data.

Statistical Rigor Assessment

The analysis is limited by the use of digitized mean data rather than individual patient data, precluding estimation of interindividual variability and formal hypothesis testing. No VPC, bootstrap, or cross-validation was performed. Parameter uncertainty was explored through scenario analyses ('bookend' efficiencies of 0.5-fold and 4-fold) and a global sensitivity analysis (Sobol method), which showed that the model is sensitive to all estimated parameters and that the timing of re-treatment is critical. The absence of formal statistical inference (e.g., confidence intervals on dose predictions) reduces the rigor, but is understandable given the limited data and the exploratory, decision-support nature of the work.

▯ Key Findings

The calibrated IS/ID model accurately reproduced the mean IDO- and PD-L1-specific T-cell responses in patients with complete response (CR) and partial response (PR) from the MM1636 trial (Figure 2). The time to maximal response was approximately 10 weeks for all antigen–phenotype combinations. Dose-equivalence analysis, based on molecular weight and expression yield assumptions, estimated that 180 μg of mRNA-4359 is equivalent to 200 μg of the peptide vaccine, with a range of 45–360 μg (4-fold higher to 2-fold lower efficiency). Model simulations of steady-state peak and trough T-cell responses (Figure 4) indicated that doses below 100 μg lie on the steep portion of the dose–response curve, whereas 100 μg is near the inflection point, making it a suitable FIH starting dose. A 15-cycle Q3W regimen was predicted to provide the longest duration of immune response (42.5 weeks) compared with Q2W or hybrid regimens (Figure 5).

💡 Clinical & Regulatory Implications

The IS/ID model predicts that a 100 μg starting dose of mRNA-4359 will elicit measurable antigen-specific T-cell responses while avoiding subtherapeutic dosing. The simulated 15-cycle Q3W regimen maximizes duration of immune response (42.5 weeks vs. 37.5 weeks for a Q2W/Q3W hybrid). These findings directly informed the dose-escalation scheme of an ongoing FIH trial (NCT05533697) and will be refined with emerging clinical data. The approach reduces the number of patients exposed to potentially ineffective doses and supports combination with pembrolizumab.

Strengths & Limitations

Strengths

- Innovative use of IS/ID modeling to support FIH dose selection in the complete absence of preclinical in vivo animal data.
- Clear, hypothesis-driven adaptation of an existing model structure to a new therapeutic modality and antigen system.
- Integration of published clinical data from a related peptide vaccine with prior platform knowledge to bridge dose equivalence.
- Practical sensitivity analysis (Sobol) and scenario testing (0.5–4 fold efficiency) to quantify uncertainty.
- The work directly informed an ongoing phase 1/2 clinical trial, demonstrating real-world impact of MIDD.

Limitations (Acknowledged by Authors)

- Only T-cell responses were modeled; responses of other immune cells (e.g., Tregs) were not considered due to lack of data.
- The model could not distinguish between CD4+ and CD8+ T-cell contributions.
- The model could not predict across different cancer indications.
- Different parameterizations were used for IDO and PD-L1; no interaction between the two antigens was assumed.
- Concomitant anti–PD-1 treatment (nivolumab) was assumed to have no effect on immune modulation.
- Model focused on mean response; future work should incorporate virtual populations to capture heterogeneity.

Limitations (Expert Review)

- The sigmoidal dose–response shape was assumed based on prior proprietary data rather than estimated from the peptide vaccine data, potentially introducing bias.
- No formal parameter uncertainty propagation (e.g., bootstrap) was performed on the dose predictions; the 'bookend' scenario analysis is a coarse sensitivity check.
- The model calibration relied on digitized mean data, which ignores within-study variability and may obscure important subgroup effects.
- The assumption that all mRNA molecules are taken up by APCs with equal efficiency is a strong simplification; in reality, LNP uptake and endosomal escape vary.
- The molecular weight and expression yield assumptions (17 proteins per mRNA, 4 IDO + 4 PD-L1 sequences) are based on unpublished data and cannot be independently verified.

Generalizability

The model was calibrated to a single trial in metastatic melanoma; generalizability to other cancers and to combination with pembrolizumab vs. nivolumab is uncertain because of potential differences in PD-1/PD-L1 axis modulation. The approach, however, is generalizable to other mRNA-based immunotherapies where a suitable comparator clinical dataset exists.

Key Equations

TEM cell dynamics

$$\frac{dT_{TEM}}{dt} = \delta(t) - \beta_{TEM} \cdot T_{TEM} - \mu_{TEM} \cdot T_{TEM}$$

Rate of change of transitional effector memory (TEM) T-cell count. TEM cells are activated from naive T cells (via the activation function δ) and convert to central memory (CM) cells at rate β_TEM; they die at rate μ_TEM.

CM cell dynamics

$$\frac{dT_{CM}}{dt} = \beta_{TEM} \cdot T_{TEM} - \beta_{CM} \cdot T_{CM} - \mu_{CM} \cdot T_{CM} + \text{Replication term (upon re-treatment)}$$

Rate of change of central memory (CM) T-cell count. On re-treatment, CM cells replicate at rate R_CM for a period τ and convert back to TEM cells at rate β_CM; they die at rate μ_CM.

Dose-dependent activation function

$$\delta(D) = \frac{E_{max} \cdot D^{\gamma}}{ED_{50}^{\gamma} + D^{\gamma}}$$

Dose-dependent activation function δ, following a sigmoidal relationship with dose D. E_max is maximal activation, ED_50 is half-effective dose, and γ is the Hill coefficient.

Dose-dependent TEM-to-CM conversion rate

$$\beta_{TEM}(D) = \beta_{TEM,\,\max} \cdot \frac{D^{\gamma_\beta}}{ED_{50,\beta}^{\gamma_\beta} + D^{\gamma_\beta}}$$

Dose-dependent conversion rate of TEM to CM cells, also following a sigmoidal relationship. β_TEM,max is the maximum conversion rate, and ED_50,β is the half-effective dose.

Figures & Tables

- **Figure 1**: Schematic of the IS/ID model showing TEM and CM cell compartments, activation (δ), conversion (β_{TEM}, β_{CM}), and death rates (μ_{TEM}, μ_{CM}).
  - *Significance*: Provides the structural framework of the semi-mechanistic model, illustrating the key dynamic processes of T-cell activation, memory conversion, and apoptosis.
- **Figure 2**: Calibration of the modified IS/ID model to mean observed IDO- and PD-L1-specific T-cell response data from the MM1636 study, stratified by complete response (CR) and partial response (PR).
  - *Significance*: Demonstrates the model's ability to capture the time course of immune response over 50 weeks, validating the model structure before dose predictions.
- **Figure 3**: Predicted dose–effect relationships for (A) the peptide-based vaccine and (B) mRNA-4359, showing sigmoidal curves with a steep rise between 10–100 µg.
  - *Significance*: Forms the basis for dose equivalence calculations; illustrates that 180 µg mRNA-4359 corresponds to the same effect as 200 µg peptide vaccine.
- **Figure 4**: Steady-state peak and trough T-cell responses as a function of mRNA-4359 dose (1–1000 µg), shown separately for IDO CR, IDO PR, PD-L1 CR, and PD-L1 PR.
  - *Significance*: Supports starting dose selection by showing that doses below 100 µg fall in the steep portion of the curve, while 100 µg is near the inflection point, enabling dose exploration.
- **Figure 5**: Simulation of three alternative mRNA-4359 dosing regimens: 15 Q2W, 15 Q3W, and 6 Q2W + 9 Q3W, showing T-cell response over 80 weeks.
  - *Significance*: Guides regimen selection by demonstrating that a 15-cycle Q3W regimen provides the longest duration of immune response (~42.5 weeks) while achieving similar peak magnitude.

Code & Reproducibility Assessment

Executable model files (SBML, MATLAB scripts) are provided as Supplementary Information, enabling independent reproduction of simulations and dose–response curves.

Supplementary Materials

Supplementary materials include: Table S1 (parameter constraints and values), Table S2 (dose scenario summaries), Figure S1 (Sobol sensitivity analysis plots), Figure S2 (activation function schematic), Figure S3 (memory cell replication schematic), Data S2 (model equations in SBML), and executable MATLAB scripts. These enable independent model reproduction.

Future Directions

Once clinical data from the FIH trial become available, the IS/ID model can be refined by (1) incorporating individual-level PK/PD data rather than mean trajectories, (2) adding a tumor compartment to link T-cell responses with tumor burden dynamics, (3) integrating covariates such as baseline MHC expression or prior checkpoint inhibitor exposure, and (4) extending to other cancer indications. The approach could be generalized to other mRNA-based immunotherapies encoding multiple antigens, potentially enabling platform-level dose–response libraries.

Expert Commentary

This paper exemplifies a pragmatic fusion of quantitative pharmacology and translational science. The team wisely leveraged a well-characterized peptide vaccine dataset to anchor a mechanistic IS/ID model, bypassing the common pitfall of over-reliance on non-human primate data that often cannot recapitulate human T-cell responses. The assumption of sigmoidal dose-effect relationships based on prior mRNA platform data (rather than fitted from the trial) is both a strength and a vulnerability—it adds prior information but constrains the model to a predetermined shape. The decision to stratify by complete vs. partial response rather than model covariates is a sensible compromise given limited data points. For the field, this work underscores that semi-mechanistic models, when combined with robust clinical data from related modalities, can provide actionable dose predictions for novel platforms. The real test will come when the FIH data become available, and the model predictions can be compared with observed responses.

Bottom Line

This analysis demonstrates that an IS/ID model calibrated to a peptide vaccine can support FIH dose selection for an mRNA-based immunotherapy without preclinical animal data. The 100 μg starting dose of mRNA-4359 balances the need to be therapeutically relevant (above the steep portion of the dose–response curve) while maintaining an adequate safety margin. For pharmacometricians, this work highlights the value of mechanism-based modeling to bridge across therapeutic modalities and accelerate early clinical development of novel immunotherapies.

Figures

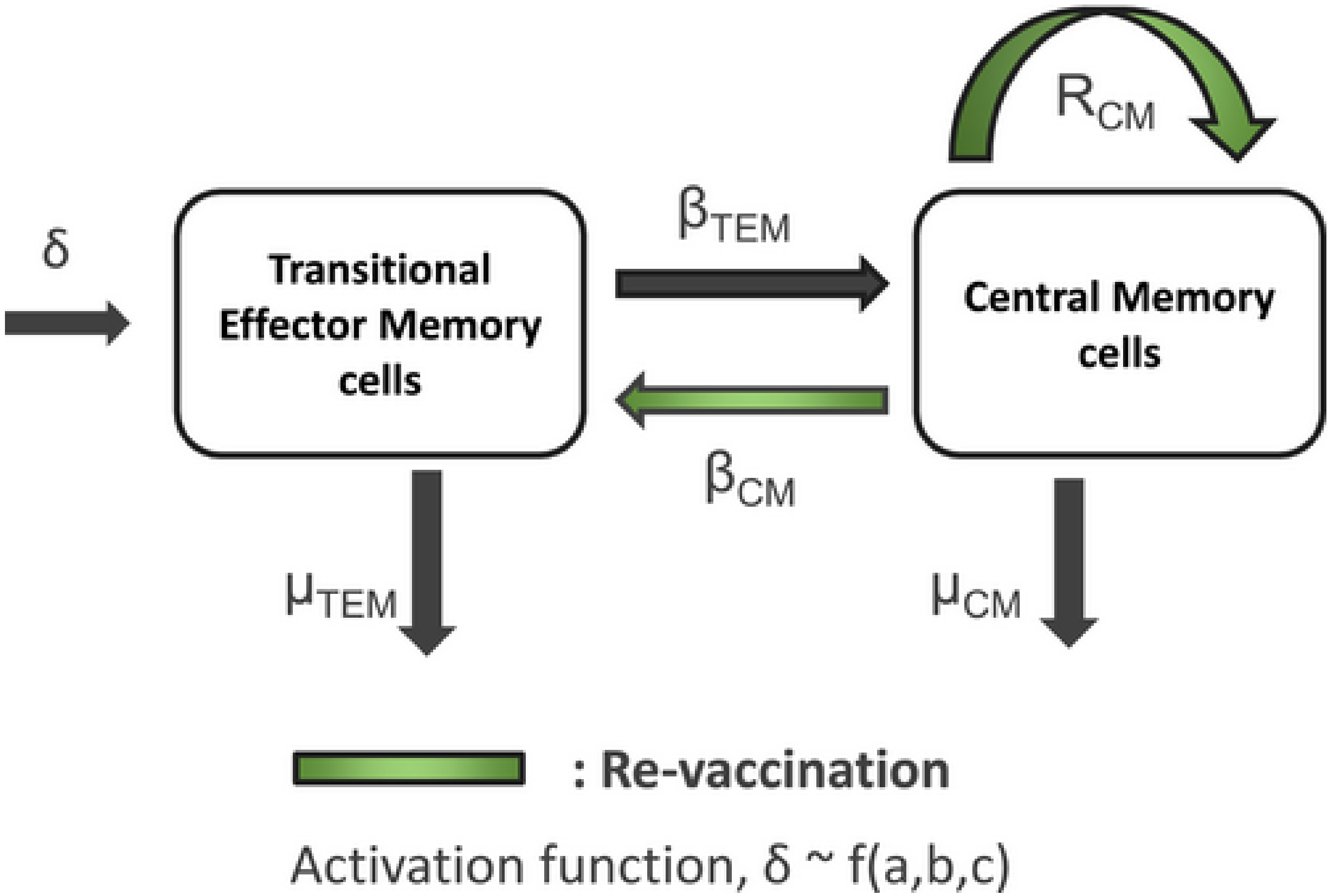

Figure 1: Schematic of the IS/ID model.

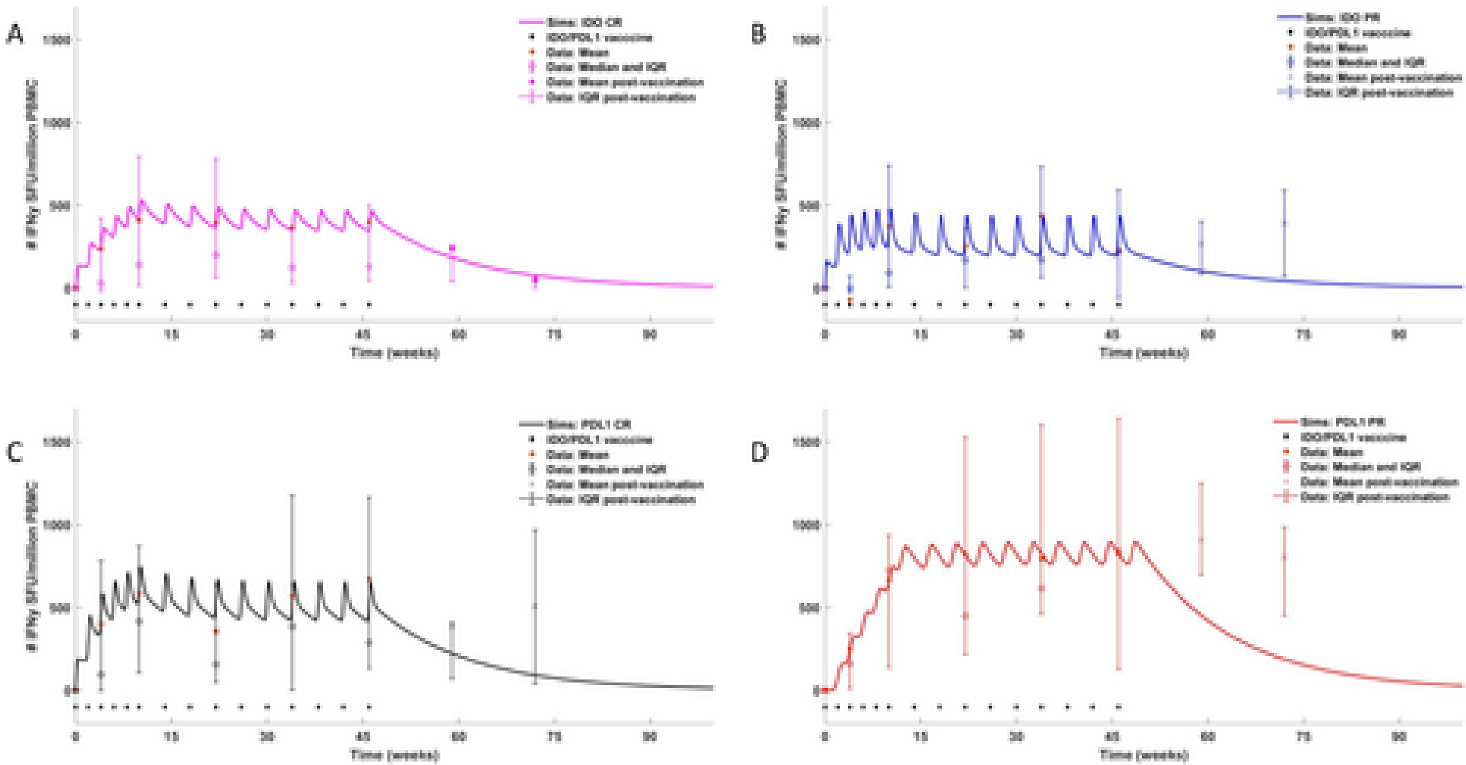

Figure 2: The IS/ID model was calibrated to previously published clinical data from the MM1636 study. Data were segregated by IDO and PD-L1 antigen–specific response and b

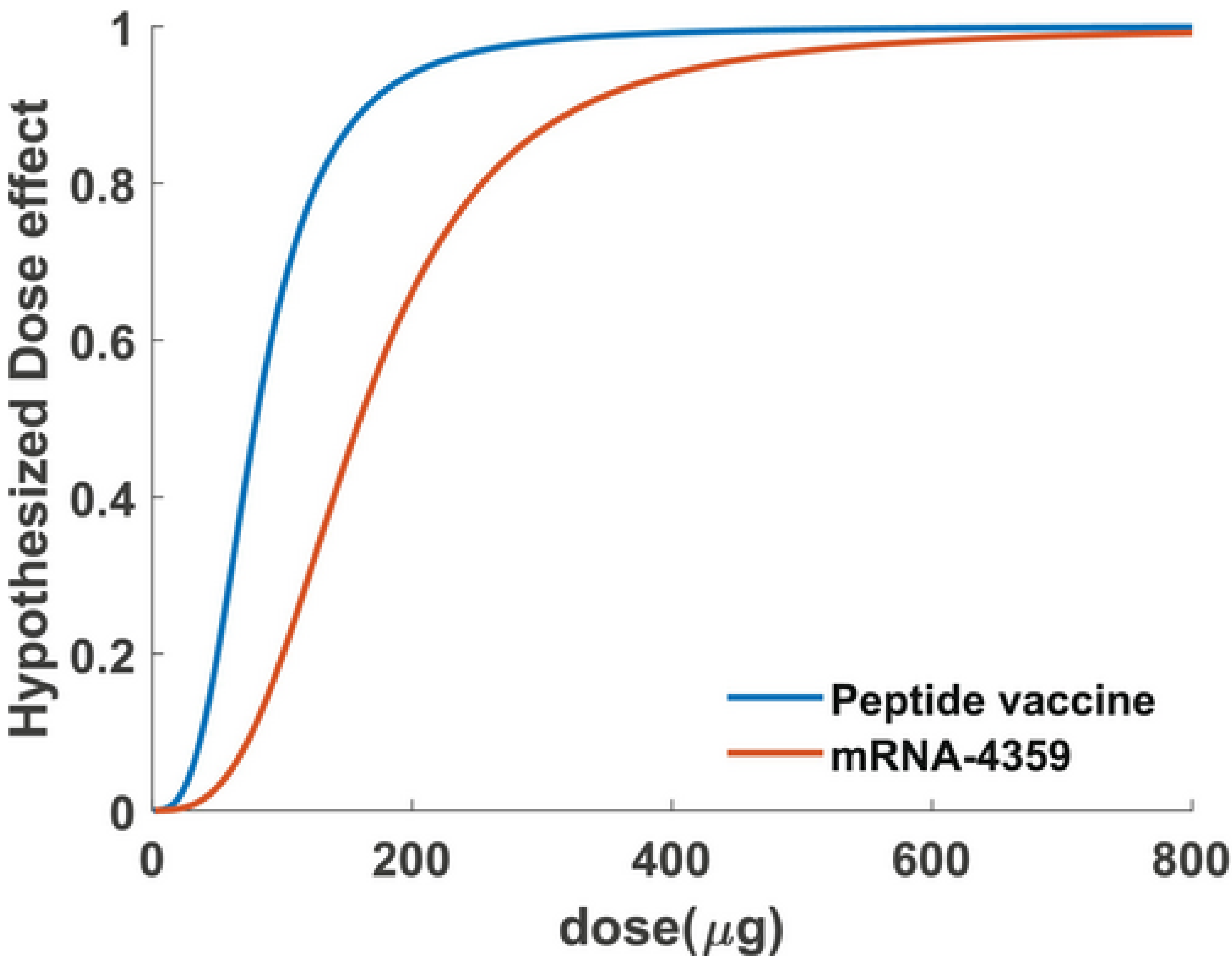

Figure 3: Predicted dose-effect relationships in the modified model for (A) the peptide-based vaccine and (B) mRNA-4359. From the dose-effect curve, the parameter values f

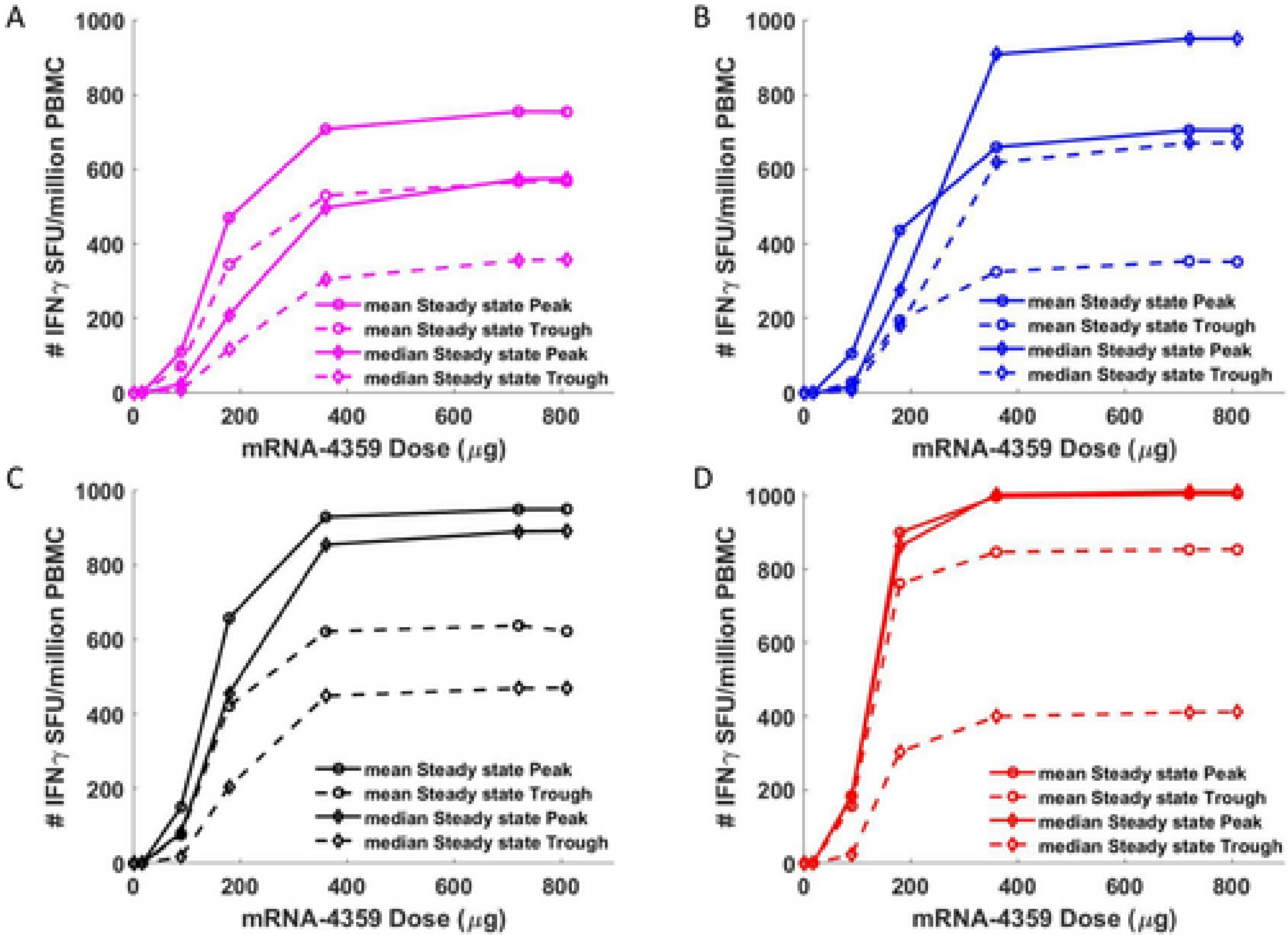

Figure 4: Dose-effect simulations according to antigen-specific response and patient RECIST 1.1 response. IDO CR (top-left), IDO PR (top-right), PD-L1 CR (bottom-left), an

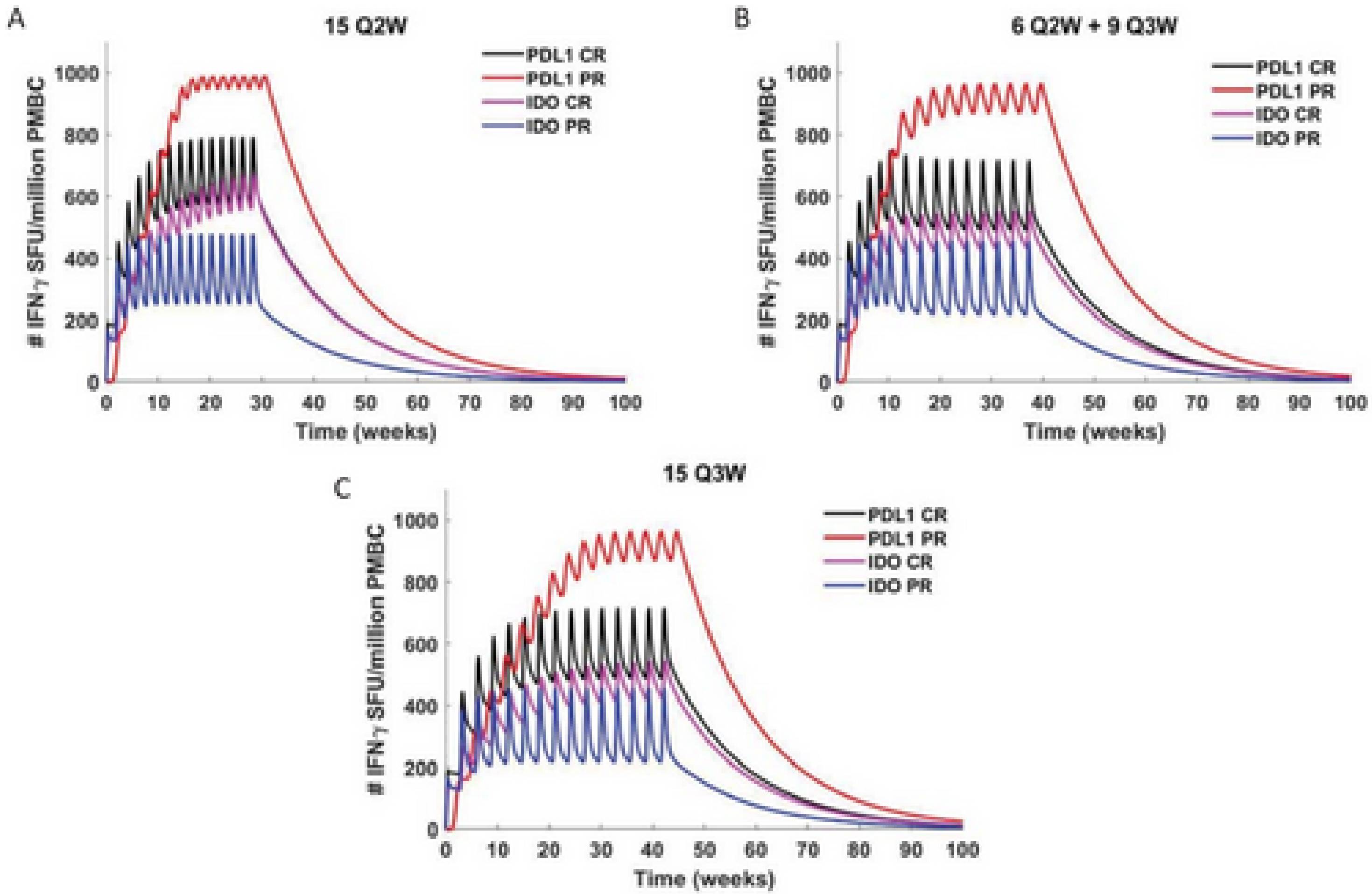

Figure 5: Model simulations for alternative dosing regimens.